

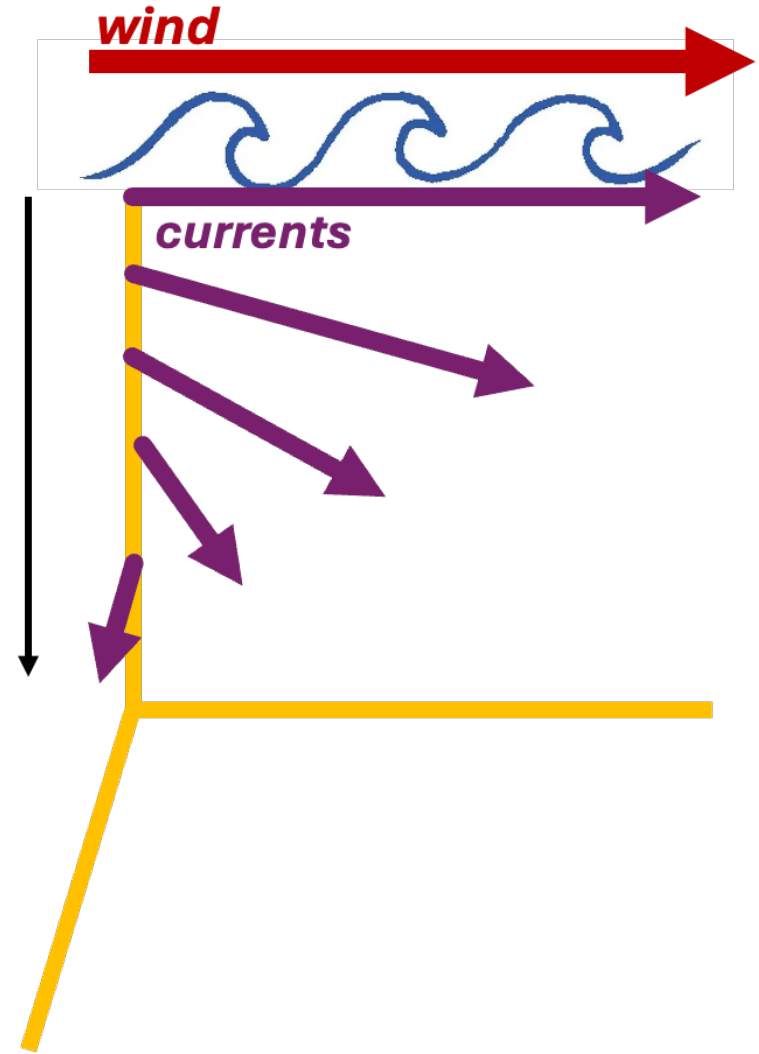
Understanding the Ekman Spiral



What are some of the atmospheric forces that move water at the surface?

How wind creates currents

- Energy is being transferred from wind to the surface and between “layers” of water through **friction**. Only a fraction of that energy is transferred.
 - With steady wind, the speed of the surface current is about 1-3% of the wind speed
 - Speed of the wind driven current decreases with depth.
- At some depth, this wind energy “runs out” and we no longer have wind-driven currents, *we still have currents – they are just not primarily driven by wind*
 - Wind driven currents are restricted to the upper 100 to 200 meters in the open ocean (even shallower in some areas)



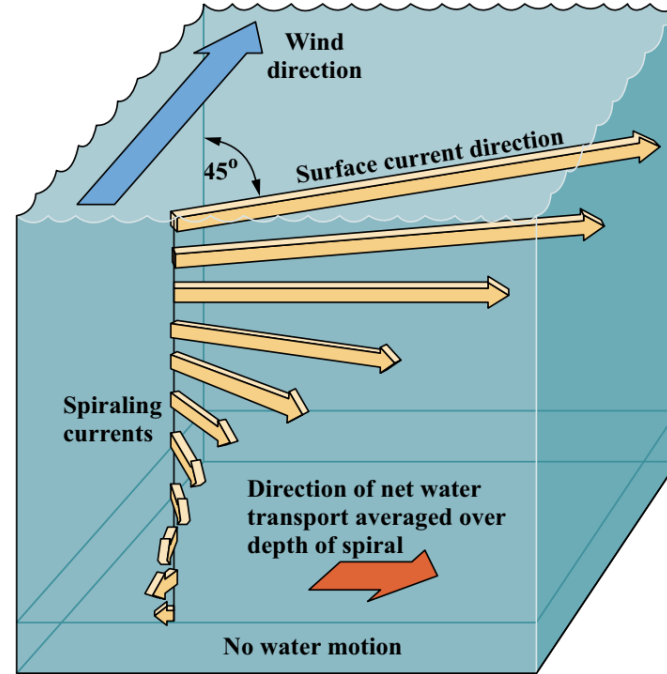
The wind driven current layer

The wind-driven current layer of the water column is called **the Ekman layer**.

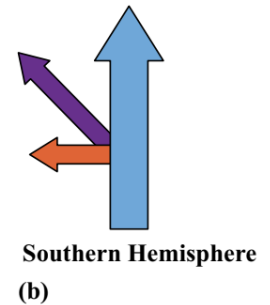
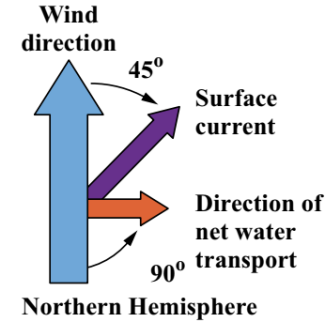
Coriolis Force causes rotation from wind to sea surface AND with currents with depth.

Ekman Theory States:

- Predicts the surface current angle to be 45° from the wind.
- The mean movement of water summed over all depths in this wind-driven layer is 90° to the wind. This phenomenon is called: **Ekman Transport**.



(a)



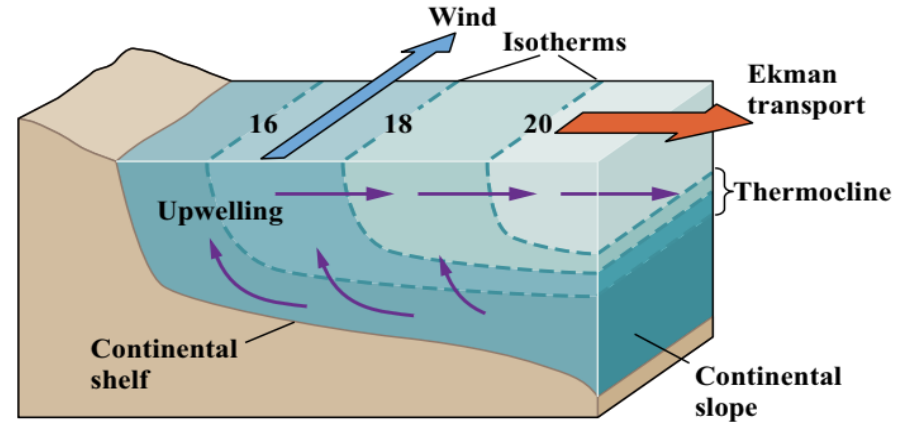
(b)

Even more wind driven
water movement

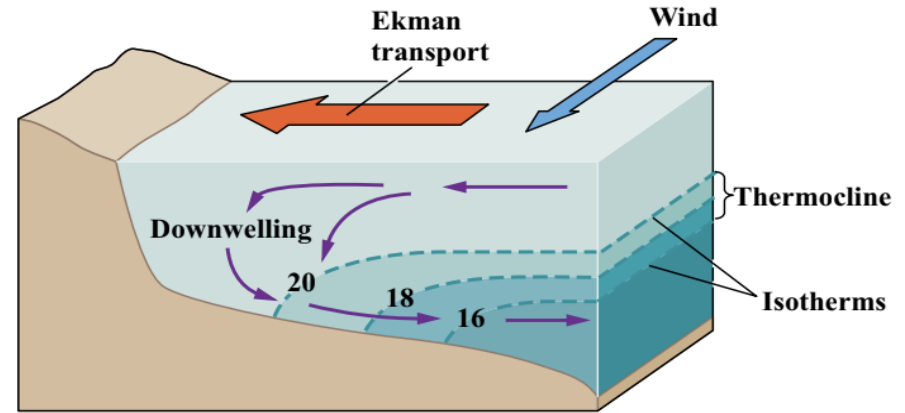
Although wind-driven surface water motions are primarily horizontal, winds can also cause vertical motion of the water – called **upwelling** and **downwelling**.

Upwelling — near the coast, Ekman transport pulls water away from land causing water to be pushed upward.

Downwelling – Ekman transport pushes water towards land, causing water to be pushed downward.



(a) Upwelling

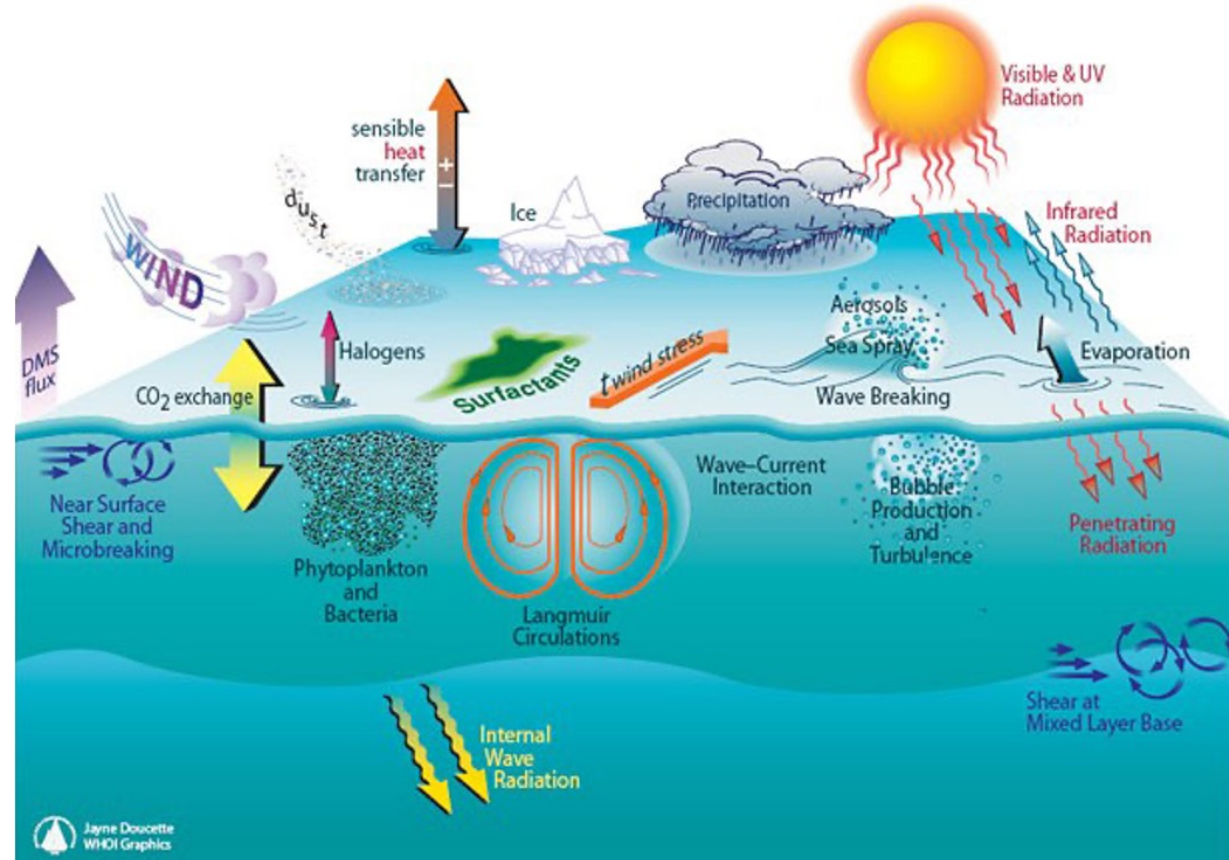


(b) Downwelling

Water mass movements →

This all seems straightforward, why do scientists still research this?

- The Ekman Spiral is a theoretical concept and difficult to observe and model directly.
- The upper ocean is complex, with many interacting physical processes.
- Understanding these upper ocean currents helps improve our knowledge of how materials like plankton, plastic, oil, and search and rescue move at the sea surface.

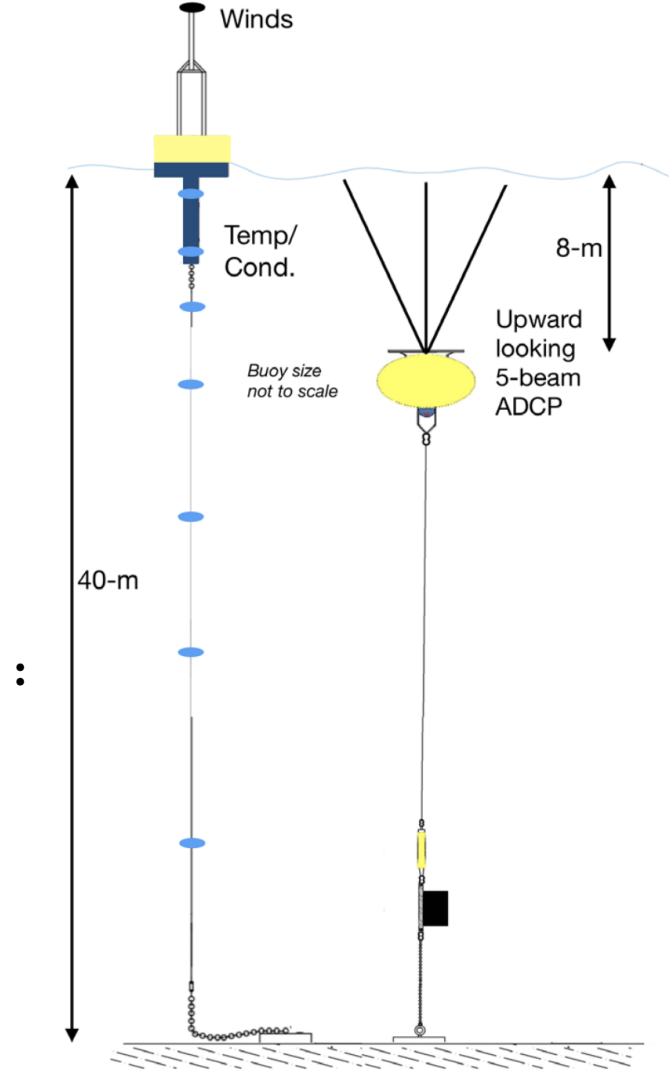


Moving to my research:

*So, we want to know more about these currents and the Ekman Spiral:
How do we measure them?*

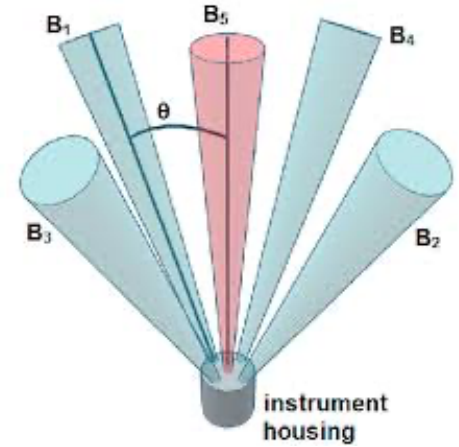
My research project utilizes Acoustic Doppler Current Profilers (ADCPs) and Shore-based High Frequency Radar (HFR)

- **ADCPs:**
 - Current measurements from 1.5 - 4m below sea surface
 - ADCPs work by transmitting sound waves ("pings") into the water and measuring the frequency shift of the sound waves that reflect off particles moving with the water



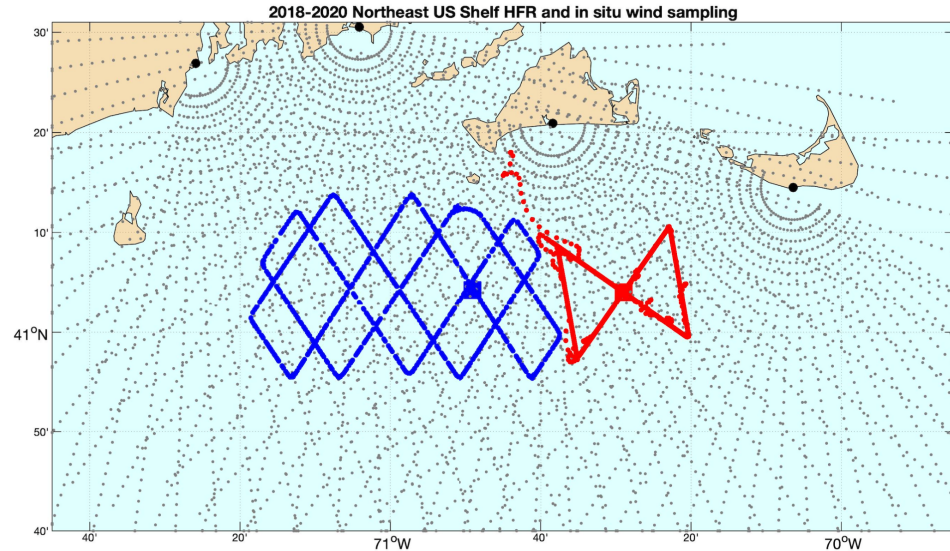
More on ADCPs:

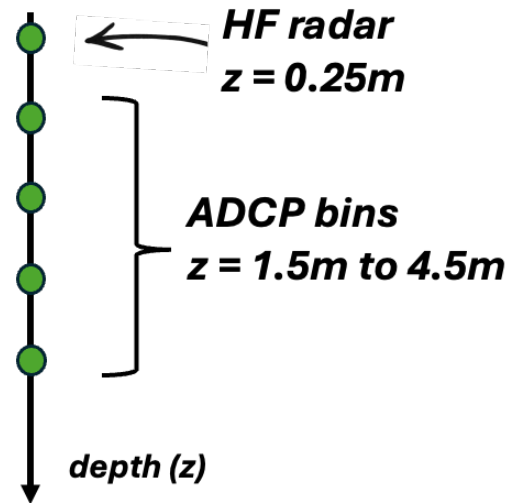
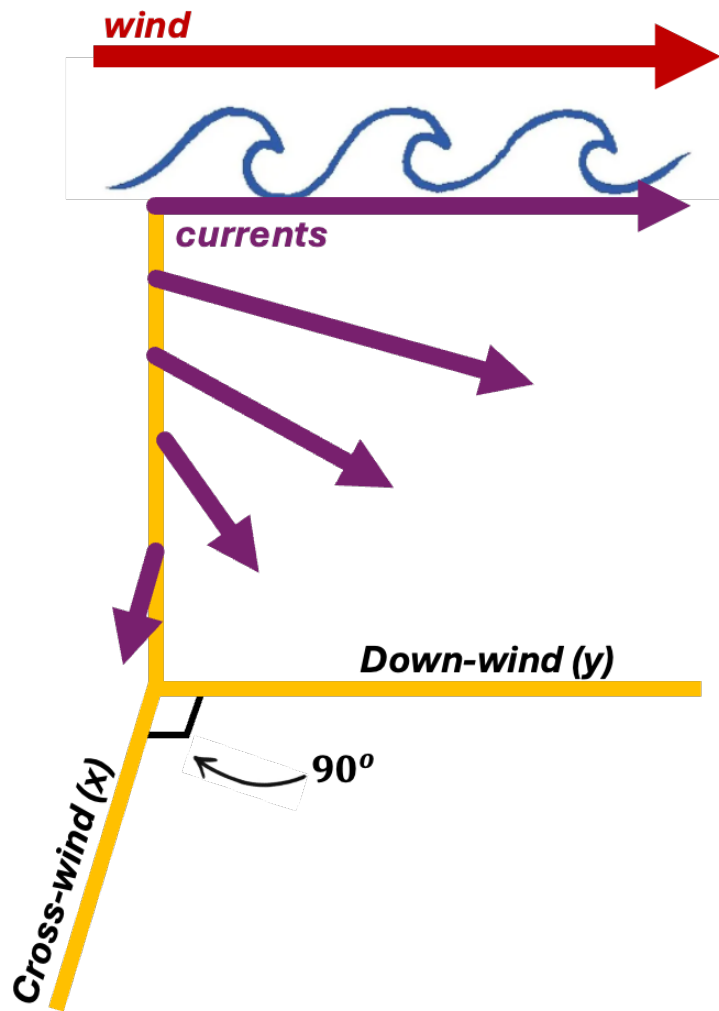
- ADCPs send sound pulses into the water and listen for echoes reflected off particles in the water
- The frequency shift (Doppler shift) in the returning sound tells how fast the water is moving toward or away from the instrument.
- These sound pulses can measure these shifts at a range of depths and in combination with the beams pointing in different directions: we are able to see currents throughout the water column.



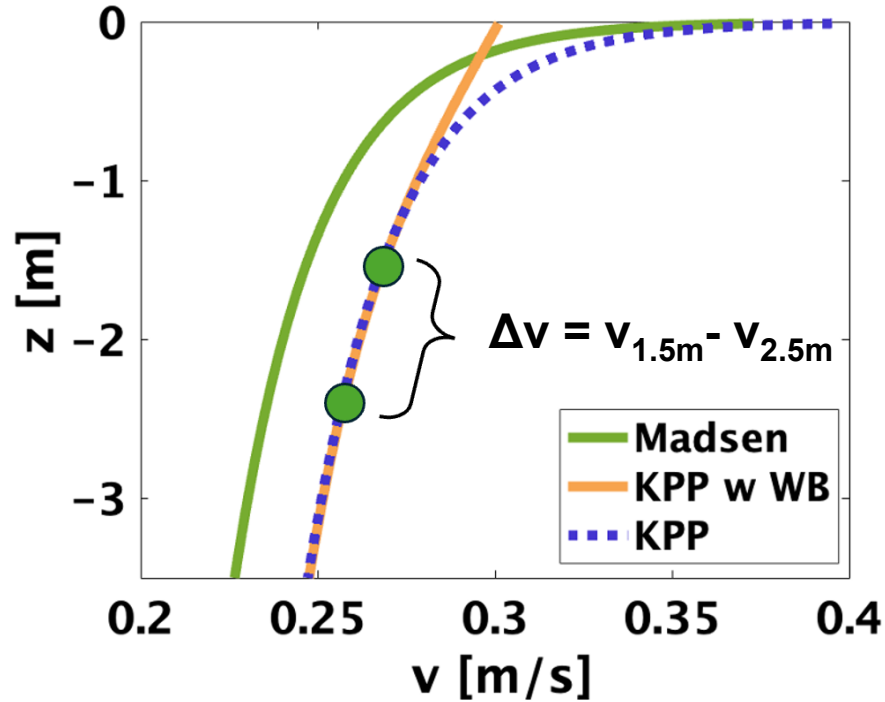
Shore Based High Frequency Radars (HFRs)

- Measures currents at an effective depth of 0.25m
- High Frequency radar (HFR) sends radio waves over the ocean surface.
- It measures how the waves bounce back to find the speed and direction of surface currents.





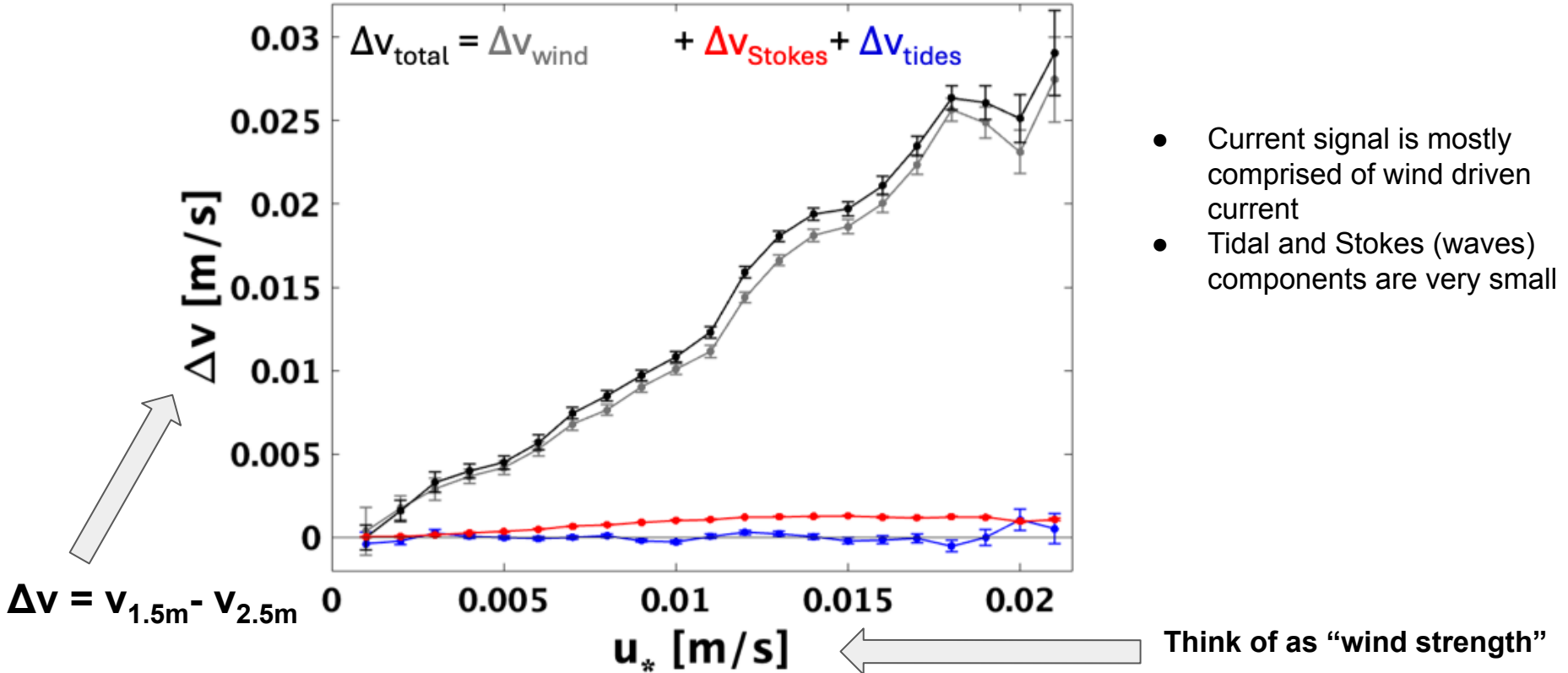
Analysis Approach

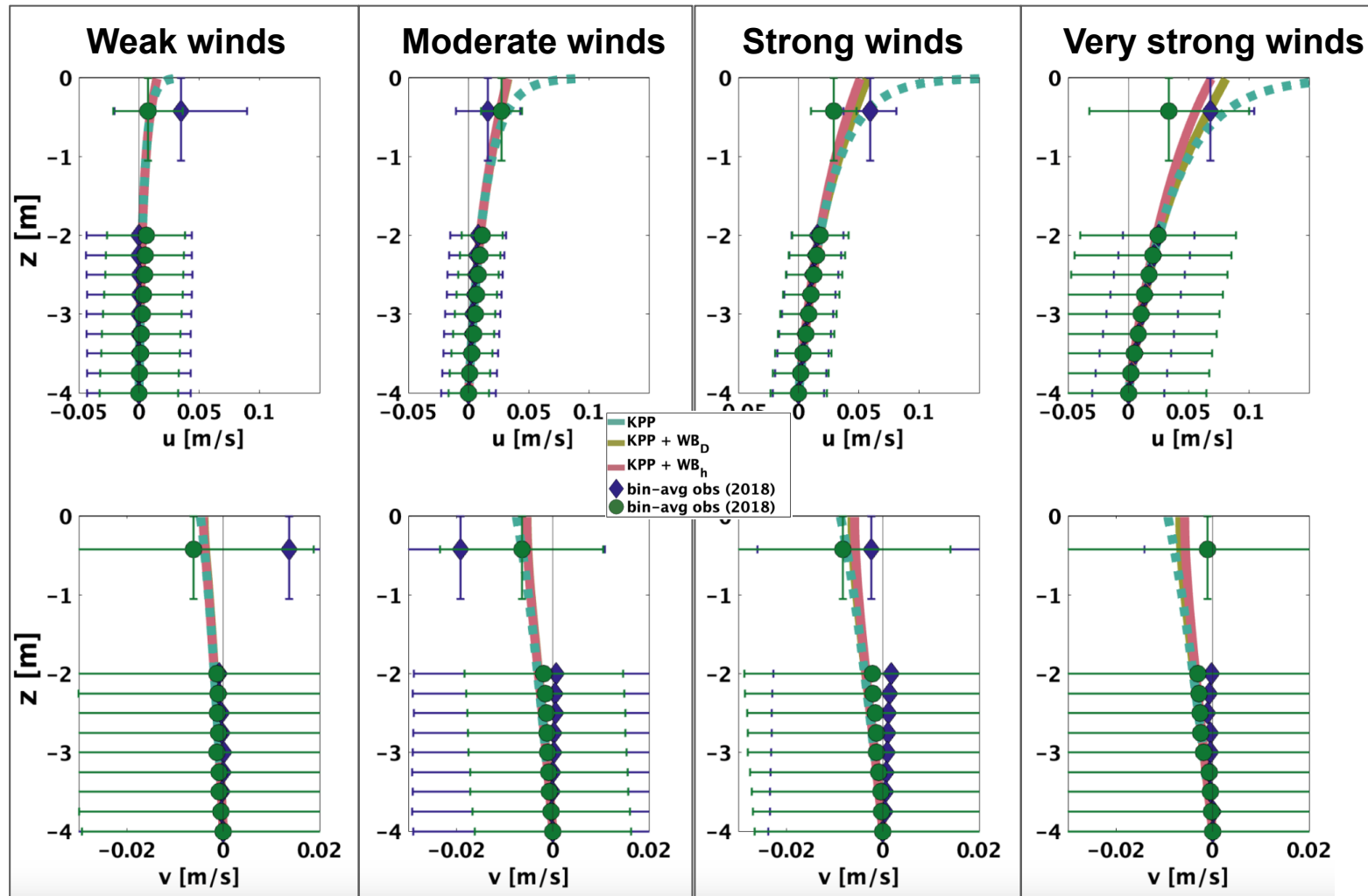


- We look at the velocity differences at different depths and plot them against u_* (which you can think of as strength of wind)
- We then compare those observations to models to understand what processes are important in predicting current speeds

Results!

Proof of large wind driven current seen in my data:





Summary of results

- We've been able to quantify how much of the total current structure is due to wind-driven currents (in the coastal ocean)
- Plotting models of the currents with our observations has revealed that **wave-breaking** may affect the currents in the upper 2m of the ocean.